

Question				Marking details				Marks Available					
								AO1	AO2	AO3	Total	Maths	Prac
6. FT	(a)	(i)		H ₃ C-N=C=O					5		5	5	
				Bond type	Number of bonds	Energy to break one bond (kJ/mol)	Energy to break all of each bond type (kJ/mol)						
				H-C	3	413	1 239 (1)						
				C-N	1	305	305						
				N=C	1	615	615 (1)						
				C=O	1	745 (1)	745						
				H ₂ O									
				Bond type	Number of bonds	Energy to break one bond (kJ/mol)	Energy to break all of each bond type (kJ/mol)						
				H-O	2	467	934 (1)						
				Total energy = 3 838 kJ/mol (ecf) (1)									
		(ii)		Greater than, exothermic , increases (3 × 1)				3			3		
		(iii)		Thermal runaway occurs when an increase in temperature (1) increases the reaction rate that causes a further increase in temperature (1)				2			2		

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	(b)	(i)		4 100 units		1		1	1	
		(ii)		12		1		1	1	
		(iii)		Correct sequence of halving (ecf)(1) = 2 (1)		2		2	2	
				Question 6 total	5	9		14	9	